Model Paper - Database Management Systems

Generated: 2026-03-04 00:12:12

Q1 (20 marks)

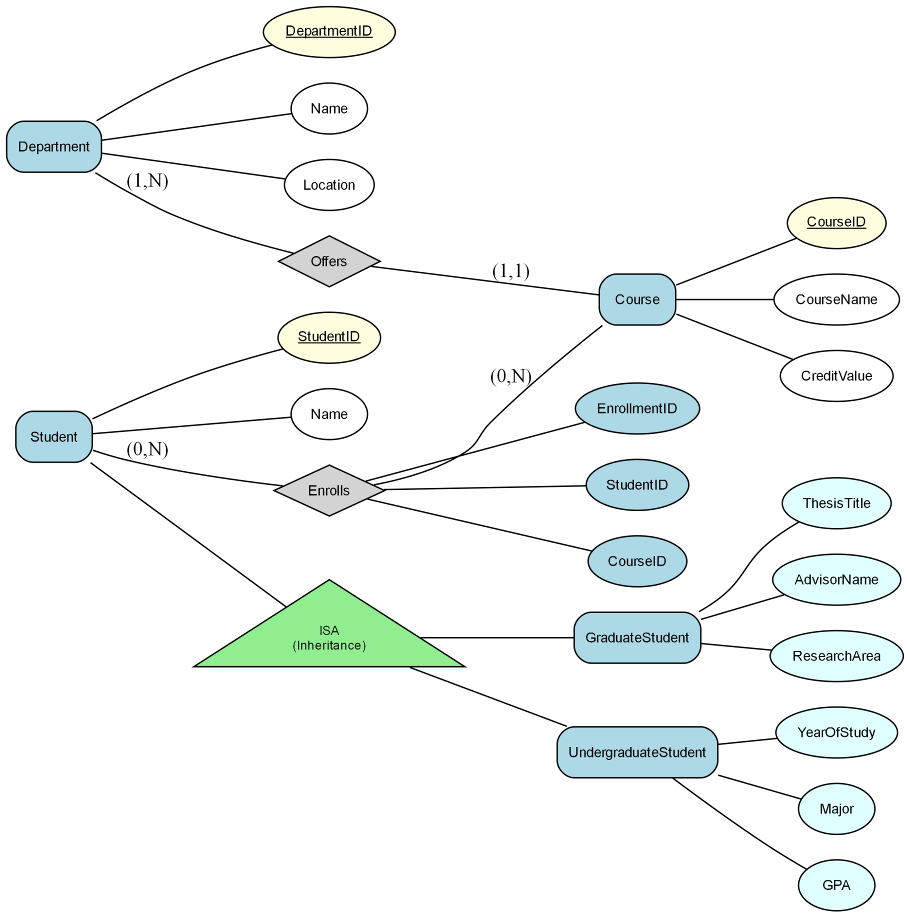
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university is managing multiple departments, each offering different courses. Each department has a unique identifier, a name, and a location. Courses are defined by a course ID, course name, and have a credit value associated. Each course is assigned to one department, but each department can offer multiple courses. Students enroll in various courses, represented by a unique enrollment ID, along with a student's ID and the course ID they are enrolled in. The university needs to effectively manage the relationships and data structure of these entities. IMPORTANT: The system includes ISA hierarchies (subtype/supertype relationships). Student is the supertype with two subtypes: GraduateStudent and UndergraduateStudent. GraduateStudent has specific attributes: ThesisTitle, AdvisorName, ResearchArea. UndergraduateStudent has specific attributes: YearOfStudy, Major, GPA. A Department can be related to many Courses through the 'Offers' relationship. A Student can be related to many Courses, and a Course can be related to many Students through the 'Enrolls' relationship (Student participation is optional, Course participation is optional)

Figure: EER



a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation R(A, B, C, D, E) with the following set of functional dependencies F over R: F={{AB->C, C->D, DE->B, A->EF}}.

a) Find all the keys in relation R using the attribute closure method. (4 marks)

b) Is R in 3NF? Give reasons for your conclusion. (3 marks)

c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A retail company is creating a database system to manage its inventory and sales records efficiently. The database schema includes tables such as Products(product_id: integer, product_name: string, price: real, stock_quantity: integer), Orders(order_id: integer, order_date: date, total_amount: real), and Order_Details(order_detail_id: integer, order_id: integer, product_id: integer, quantity: integer). The company intends to implement various database operations to maintain data integrity and support business transactions.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment:

```java
String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql);
pstmt.setString(1, "IT");
ResultSet rs = pstmt.executeQuery();
```

Which type of JDBC statement is used in the code segment shown above?
Briefly explain when this type of statement will be used. (2 marks)

c) Briefly explain the options available for transferring data between two sources such as between tables and servers. (2 marks)

d) Briefly explain how authentication and authorization are achieved in SQL Server. (3 marks)

e) An organization is establishing role-based access control for its database system. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the operational database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating transaction data.

i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)
ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)
iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the operational database using a user-defined server role. (5 marks)
iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)
v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the operational databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for a hospital: Patient (patientId: int, name: varchar(100), dob: date, address: varchar(150), phone: varchar(15)) Appointment (appointmentId: int, patientId: int, appointmentDate: date, doctorId: int, status: varchar(50)) Doctor (doctorId: int, name: varchar(100), specialty: varchar(50), phone: varchar(15)) Fee (feeId: int, appointmentId: int, amount: real, paymentStatus: varchar(50)) The 'Patient' table stores patient information, the 'Appointment' table details their appointments, the 'Doctor' table contains doctors' information including their specialties, and the 'Fee' table tracks appointment charges and payment statuses. The 'Appointment' table holds appointment records, tracking scheduled appointments between patients and doctors with dates and times. The 'Fee' table contains relevant information for the database system.

a) Write SQL Queries to perform the following:

i. Find the names and phone numbers of patients who have appointments scheduled for '2023-10-15'. (4 marks)

ii. Find the patient who has the highest total amount compared to all other patients. Find the patientId and name. (6 marks)

iii. Find the appointmentDate, status, and name from Appointment, Patient, and Appointment where Appointment.appointmentDate is not null. (7 marks)

b) b) Create a function to calculate the total fee amount for a given patient. This function should account for fee with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'amount' column in the 'Fee' table whenever a new fee record is added, or an existing fee record is updated. The 'amount' column should contain the total fee amount for each patient. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by patient. The patient pays only 50% of the total fee amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)